

ZOO201

Current Objective and Subjective Questions

2021

Objective Part

- ❖ The pheromones that are used to warn other individuals of danger are **Alarm pheromones**
- ❖ In insects that display ametabolous metamorphosis, the primary difference between adults and larva are **body size** and **sexual maturity**
- ❖ In social insects reproductive females are referred as **Queen**
- ❖ Paurometabolous metamorphosis, immature are called **Nymphs**
- ❖ Water vascular system in echinoderms is also called **Aquiferous system**.
- ❖ Which of the following shows **paurometabolous metamorphosis**.
 - a) **Grasshoppers** b) Silverfish c) Butterflies d) Honeybees
- Ecdysteroids are often called as **molting hormones**
- A tube feet often has a **suction cup** at its distal end.
- The echinoderms skeletal consists of series of calcium carbonate plates called **ossicles**.
- Which organ is involved in gaseous exchange in star fish? **Dermal branchiae**
- Immature stage in hemimetabolous metamorphosis? **Nymph**
- Which of the following shows ametabolous metamorphosis?
 - a) **Silverfish** b) Butterfly c) Moth d) Cockroach
- In asteroids the larger, oral stomach, sometimes called the **cardiac stomach**.
- Each kind of individual in an insect colony is called a **caste**.
- Which of the following are exclusively marine animals?
 - a) Molluscs b) **Echinoderms** c) cnidarian d) fishes
- **Star fish** make up class asteroidae.
- Larva of sea star is known as **bipinnaria larva**.
- Complete metamorphosis consists of **4** stages.
- Member of class Ophiuroidea among following is?
 - a. Sea star b. Sea cucumber c. **Brittle star** d. Sea urchin
- An insect that transmit a disease is called **parasite**.
- ❖ Each colony can contain about **2 million zooids**.
- ❖ Most ectoprocts are **monoecism**.
- ❖ Echinoids are **dioeciously**.

- ❖ Sea cucumbers are **dioeciously**.
- ❖ Ectoprocts grow by **budding**
- ❖ All species are aquatic (both freshwater and marine) and less **than 1.5 mm** in length.
- ❖ Brachiopods are commonly called **lampshells** because they resemble **ancient Roman oil lamps**.
- ❖ An anus always open outside of the **lophophore**.
- ❖ **Metamorphosis** occurs after larvae attach to the substrate.
- ❖ The unattached end of a sea lily is called the **crown**.
- ❖ Sea lilies attach permanently to their substrate by a **stalk**.
- ❖ Members of the class **Crinoidea** include the **sea lilies and the feather stars**.
- ❖ Sea cucumbers have **no true brain**.
- ❖ Most sea cucumbers ingest particulate organic matter using their **tentacles**.
- ❖ The class Holothuroidea has approximately **1,500 species**, whose members are commonly **called sea cucumbers**.
- ❖ A chewing apparatus, **called Aristotle's lantern**, can be projected from the mouth.
- ❖ The digestive tract forms a U-shaped gut in the **calyx**.
- ❖ There is no larval stage or metamorphosis in **Chaetognaths**.
- ❖ Sea cucumbers can Evert tubules of the respiratory tree, called **Cuerverian tubules**.
- ❖ A **pheromone** that the queen releases controls the honeybee caste system
- ❖ The body of an insect is divided into three tagmata: **head, thorax, and abdomen**.
- ❖ An **insect growth regulator (IGR)** is a substance (chemical) that inhibits the life cycle of an insect.
- ❖ Grasshopper tympanal organs can detect sounds in the range of **1,000 to 50,000 cps**.
- ❖ The human ear can detect sounds between **20 and 20,000 cps**.
- ❖ **Thermoregulation** is a requirement for flying insects.
- ❖ Arrow worms are active **predators**.
- ❖ Arrow worms are active **swimmers**.
- ❖ Most Chaetognaths are planktonic, **0.5 to 12 cm** in length,
- ❖ Entoprocts have a small body cavity that most zoologists consider a **pseudo coelom**.
- ❖ Most entoprocts are **microscopic**.
- ❖ Cycliophores can reproduce asexually by producing a larva stage, called the **Pandora** that is released and settles on the same host.
- ❖ In Cycliophores **Protonephridia** are used in excretion.

- ❖ The asexual feeding stage of symbion is about **0.3 mm** tall.
- ❖ The body wall contains gland cells that eventually secrete the **chitinous tube**.
- ❖ The **gonads** are in the coelom.
- ❖ A flap of tissue called the **epistome** covers the mouth.
- ❖ The Phoronids consist of about a dozen marine species divided between two genera: **Phoronis and Phoronopsis**.
- ❖ **Statoblasts** are asexually produced and accumulate in the **metacoel**.
- ❖ A statoblast is a hard, resistant capsule containing a mass of **germinative cells**.
- ❖ Some freshwater ectoprocts produce a dormant stage called a **statoblast**.
- ❖ In others, simple gonads produce both sperm and eggs in the same **autozoid**.
- ❖ A pair of tubes called **respiratory trees** attach at the rectum and branch throughout the body cavity of **sea cucumber**.
- ❖ A **pedicellaria** is a small wrench-or claw shaped appendages with movable jaws, called valves.
- ❖ Zoologists believe that echinoderms show **common ancestry** with hemichordates and chordates because of **deuterostome characteristics** that they shows.

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Subjective Part

Q. Skeleton of sea urchin? Explain

Sea urchins are rounded, and their oral end is oriented toward the substrate.

- Their skeleton, called a test, consists of 10 closely fitting plates that arch between oral and amoral ends. Five rows of ambulacral plates have openings for tube feet, and alternate with five inter ambulacral plates, which have tubercles for the articulation of spines. The base of each spine is a concave socket, and muscles at its base move the spine.

- Spines are often sharp and sometimes hollow, and they may contain venom dangerous to swimmers.

Q. Garden centipedes.

Members of the class Symphyla (**sim-filah**) (Gr. **sym**, **same** **phyllos**, **leaf**) are small arthropods (2 to 10 mm in length) that occupy soil and leaf mold, superficially resemble centipedes, and are often called garden centipedes.

Q. Phylum of sea urchin?

Sea urchin is invertebrates and belong to phylum Echinoderm.

Q. Circulatory system?

The circulatory system of insects is similar to that described for other arthropods, although the blood vessels are less well developed. Insects, like all other arthropods, have an open circulatory system which differs in both structure and function from the closed circulatory system found in humans and other vertebrates. In a closed system, blood is always contained within vessels (arteries, veins, capillaries, or the heart itself).

Q. Asexual process in chilophores

Although fragmentation, fission, and budding are methods of asexual reproduction in a number of fungi, the majority reproduce asexually by the formation of spores. Spores that are produced asexually are often termed mitospores, and such spores are produced in a variety of ways.

Q. Insects pollination? Long

Insects are responsible for the pollination of approximately 65% of all plant species.

- Insects and flowering plants have co evolutionary relationships that directly benefit humans. The annual value of insect-pollinated crops is estimated at \$19 billion per year in the United States. Insect pollination is crucial to most gardens and is as simple as insects like bees, butterflies and wasps flying from flower to flower in order to collect nectar.

- In the process, pollen collects on their bodies and rubs off on other flowers that they visit.

Q .Name of animal of diplopoda?

Members of the class diplopods are Millipedes Most millipedes are slow-moving detritivores, eating decaying leaves and other dead plant matter

Q. Trepang?

The body wall of sea cucumbers, when boiled and dried, is known as trepang in Asian countries.

Q. Starfish stomach:

The mouth opens to a short esophagus and then to a large stomach that fills most of the coelom of the central disk. The stomach is divided into two regions. •The larger, oral stomach, sometimes called the cardiac stomach, receives ingested food. •It joins the smaller, aboral stomach, sometimes called the pyloric stomach.

Q. Complete metamorphosis?

Complete metamorphosis is the type of insect development that includes egg, larva, pupal, and adult stages, which differ greatly in morphology. The lifecycle of butterflies, ants, fleas, bees, beetles, moths, and wasps are examples of the complete metamorphosis.

Q. Ventilating process in insects? Long

Most insects have ventilating mechanisms that move air into and out of the tracheal system.

- Insects do not breathe through their mouths as we do.
- The do not have lungs and their blood, which is a watery, yellowish liquid, does not carry oxygen and carbon dioxide around their bodies.

- Insects have a system of tubes, called tracheae, instead of lungs.
- These tracheae penetrate right through the insect's body.
- Air enters the tracheae by pores called spiracles.
- These spiracles are found on each side of the insect's abdomen. Each segment of the abdomen has a pair of spiracles.
- As oxygen diffuses from the tracheae to the body tissues, and is not replaced by carbon dioxide, a vacuum is created that draws more air into the spiracles. This process is called passive suction.

Q. Juvenile hormone?

The true hormones produced by epithelial glands and belonging to the ecdysteroids or juvenile hormones.

Q. Do insects have memory?

Studies over the past century have discovered that many insects, like humans, acquire more than one type of spatial memory, that they acquire these memories at different rates and that, as they become more familiar with an environment, they change which memories they use.

Q. Sea lily and star?

PHYSICAL CHARACTERISTICS: Sea lilies have a stalk below the crown and look like flowers. Feather stars have a cluster of tentacles below the crown and look like ferns. Feather stars are white, purple, and brown, green. Or violet. Sea lilies are white.

HABITAT: Most sea lilies live in deep water and most feather stars live on coral reefs.

DIET: Both are planktonic

Q. Locomotion in Sea Cucumber?

Sea cucumbers are mostly sluggish burrowers and creepers, although some swim by undulating their bodies from side to side.

•Locomotion using tube feet is inefficient, because the tube feet are not anchored by body wall ossicles. “Locomotion more commonly results from contractions of body-wall muscles that produce wormlike, locomotor waves that pass along the length of the body. One way that sea cucumbers can confuse or harm predators is by propelling their own toxic internal organs from their bodies in the direction of an attacker. •The organs grow back, and it may save them from being eaten.

Q. Name of animals of diplopodia Millipedes.

Most millipedes are slow-moving detritivores, eating decaying leaves and other dead plant matter

Q. Synchronous Flight?

- Direct or synchronous flight is a mode of transportation that is fueled by wing muscles that insert directly into the wing base.
- The insertion point of the wing is hinged which enables the muscles downward movements to lift the wing portion upward and upward movements pull the wing portion downward.

Q. How cycliophores produce by asexually?

- It can reproduce asexually by producing a larva stage, called the Pandora that is released and settles on the same host.
- Because of this asexual reproduction, large number of symbion can build up on one host.

Q. Nervous System of Insects? Long repeated –

The central nervous system consists of a series of ganglia that supply nerves to successive segments of the body. - The three main ganglia in the head (protocerebrum, deutocerebrum, and tritocerebrum) commonly are fused to form the brain. - The rest of the ganglionic chain lies below the alimentary canal against the ventral body surface. - - In most insects the number of separate ganglia has been reduced by fusion. The last abdominal ganglion always serves several segments. - In homopterans and heteropterans all the abdominal ganglia usually fuse with mesothoracic and metathoracic ganglia

Q. Class Crinoidea:

General Features - Stem and a crown consisting of a cup-like central body known as the theca. - Five rays or arms. - Mouth surrounded by feeding arms and is linked to a u-shaped gut. - Passive suspension feeders. - Dioecious - Gonads are located in the pinnules but few in the arms.

Q. Ossicles?

Series of overlapping skeletal plates known as ossicles. The echinoderm skeleton consists of a series of calcium carbonate plates called ossicles.

Q. Alarm pheromones:

Released to warn other individuals of danger; may cause orientation toward the pheromone source and elicit a subsequent attack or flight from the source.

Q. Insect have memory?

Research has demonstrated that insects are capable of some learning and have a memory. For example, bees (order Hymenoptera) instinctively recognize flowerlike objects by their shape and ability to absorb ultraviolet light, which makes the center of the flower appear dark.

Q. Sea star regeneration?

Sea stars are well known for their powers of regeneration.

- They can regenerate any part of a broken arm.
- In a few species, an entire sea star can be regenerated from a broken arm if the arm contains a portion of the central disk. Regeneration is a slow process, taking up to a year for complete regeneration.

Q. Four stages of metamorphosis?

- Egg
- Larva
- pupa and
- Adult.

Q. Pheromones?

Pheromones are chemicals an animal releases that cause behavioral or physiological changes in another member of the same species. •Zoologists have described many different insect uses of pheromones.

Q. Define vector:

A vector is a living organism that transmits an infectious agent from an infected animal to a human or another animal. Vectors are frequently arthropods, such as mosquitoes, ticks, flies, fleas and lice

Q. What are Soil dwelling insect?

Many other insects are also beneficial. •Soil-dwelling insects play important roles in aeration, drainage, and turnover of soil, and they promote decay processes. •Other insects serve important roles in food webs. •Soil-dwelling insects are hidden in the soil and hard to detect until the sudden appearance of damage creates the demand for rapid, curative treatment.

Q. Royal jelly?

Royal jelly is a honey bee secretion that is used in the nutrition of larvae, as well as adult queens. It is secreted from the glands in the hypopharynx of nurse bees, and fed to all larvae in the colony, regardless of sex or caste.

Q. Gaseous exchange in Ophiuroids?

Ten bursa on the bottom of the central body disk. - Cilia within the sacs direct water flow so that oxygen can be absorbed from the water - Water flows through cilia or muscular contraction. - Oxygen is transported by the hemal system.

Q. Class Ophiuroidea belong to which phylum?

Phylum Echinodermata

Q. Paurometabolous or metamorphosis?

Paurometabolous is the process of gradual metamorphosis. Paurometabolous is the type of growth that occurs in insects such as cockroaches, which includes just three stages, i.e. egg, nymph, and adult. It is called incomplete metamorphosis or paurometabolous since it lacks the pupa stage.

Q. Classes of hormones in insects.

There are two main classes of hormones in insects:

- 1) The true hormones produced by epithelial glands and belonging to the ecdysteroids or juvenile hormones.

2) The neuropeptide hormones produced by neurosecretory cells.

Q. Aristotle's Lantern in Echinoids?

Conical shaped chewing apparatus projecting from mouth. - Complex arrangement of muscles and calcareous teeth constituting about 35 ossicles. - Teeth are extruded to scrape algae and other food from rocks. - Five strong teeth, 40 skeletal ossicles and 60 muscles. - Masticating the food. - Feed on algae, bryozoans, coral polyps, and dead animal remnants.

Q. Class Asteroidea?

1,900 species and second largest group in the Echinodermata.

- They often live on hard substrates in marine environments.
- Thin folds of the body wall, called dermal branchiae.
- Pincher-like structures called pedicellariae.
- Ambulacral groove and paired rows of tube feet.
- Suction disks sensory receptors are distributed over the surface of the body and tube feet.
- Sea stars are dioecious, and external fertilization is the rule.
- Bipinnaria larva

Q. What is trapping?

Insect traps are used to monitor or directly reduce populations of insects or other arthropods, by trapping individuals and killing them.

Q. Stomach of Star fishes.

A sea star has 2 stomachs, the cardiac stomach and the pyloric stomach. It can push the cardiac stomach out of its mouth, in the center of its underside, to engulf prey or insert it into prey (between 2 shells, for example). The stomach then secretes a powerful digestive enzyme to break down the prey.

Q. Mating Behavior in Insects? Long

Mating behavior: Mating behavior is typically viewed as comprising all events from pair formation through courtship to the final breakup of the mating pair. Locating and recognizing mates: - Chemical attractants - Humming sound - Courtship ritual mating: Internal fertilization by insertion of the male intromittent organ into the female genital tract for deposition of sperm is the usual method of copulation. BY

PPT: Most insects have complex mating behaviors for locating and recognizing a potential mate, for positioning a mate for copulation, or for pacifying an aggressive mate. Mating behavior may involve pheromones (moths, order Lepidoptera), visual signals (fireflies, order Coleoptera), and auditory signals (cicadas, order Homoptera; and grasshoppers, crickets, and katydids, order Orthoptera).

Q. Hemal system in echinoderm?

In echinoderms, the blood vascular system is called hemal system, which is poorly developed in star fish and plays little role in circulation. The system consists of small fluid-filled sinuschannels that lack a distinct **lining**. **The haemal channels are actually intercommunicating spaces and are not true blood vessels.**

Q. Dermal Branchiae?

External gill-like structures are scattered across their body surface. - Gas exchange. - Papulae. - Cilia beat and ventilate the respiratory surface. - Outer ectodermal epithelium, gland cells and a small basiepithelial nerve plexus. - Papulae function as a simple site of respiratory exchange.

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MUHAMMAD AFTAB & HUMA KHAN
BEST OF LUCK